

USTKA, Poland

WMO: 121150

Lat: 54.5883N

Lon: 16.8542E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-7.8	-14.1	1.1	-9.1	-11.4	1.4	-6.7	18.0	2.3	16.2	3.2	3.5	140	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	26.8	18.8	24.4	18.5	22.6	18.0	20.6	24.2	19.6	22.7	18.7	21.7	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.2	22.2	18.4	13.3	21.3	17.6	12.6	20.5	59.6	23.9	56.1	22.7	53.2	21.7	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.5	10.9	DB	-13.0	31.6	3.8	2.0	-15.7	33.0	-17.9	34.2	-20.1	35.3	-22.8	36.8	(4)
(5)			WB	-13.5	22.2	3.6	1.3	-16.1	23.2	-18.2	23.9	-20.3	24.6	-22.9	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.5	1.1	3.2	7.3	11.5	15.2	17.9	18.0	14.8	10.0	5.5	2.3	(6)
(7)		DBStd	7.08	4.35	3.85	3.42	3.65	3.49	2.80	2.55	2.50	2.55	3.07	3.23	3.99	(7)
(8)		HDD10.0	1291	294	249	212	99	21	0	0	0	1	38	137	239	(8)
(9)		HDD18.3	3480	553	482	470	333	216	102	39	36	109	258	385	497	(9)
(10)		CDD10.0	918	0	0	1	17	67	156	244	248	144	39	2	0	(10)
(11)		CDD18.3	66	0	0	0	1	3	8	24	26	3	0	0	0	(11)
(12)		CDH23.3	328	0	0	0	10	36	49	109	107	17	0	0	0	(12)
(13)		CDH26.7	81	0	0	0	2	8	13	29	27	2	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.7	5.5	5.4	5.0	4.6	4.9	4.8	4.6	4.9	5.2	5.2	5.8	(14)
(15)	Precipitation	PrecAvg	687	44	37	40	30	48	59	75	87	79	72	61	53	(15)
(16)		PrecMax	947	120	57	95	63	119	202	156	177	220	217	127	103	(16)
(17)		PrecMin	524	5	6	9	8	16	1	16	8	20	17	4	23	(17)
(18)		PrecStd	103	27	14	23	16	25	39	36	39	42	44	29	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	10.9	15.4	24.4	28.0	28.5	30.5	30.0	25.7	19.7	13.2	11.2	(19)
(20)			MCWB	8.3	8.0	9.0	14.7	17.6	18.3	20.7	20.1	17.0	14.3	10.7	9.6	(20)
(21)		2%	DB	8.4	8.8	11.5	18.8	22.5	24.0	26.0	26.0	22.6	17.1	11.7	9.5	(21)
(22)			MCWB	7.0	6.8	8.4	11.8	15.3	17.5	20.1	19.3	16.4	13.7	10.2	8.0	(22)
(23)		5%	DB	6.9	7.2	9.3	15.0	18.8	21.2	23.8	23.7	20.6	15.5	10.7	8.1	(23)
(24)			MCWB	5.8	5.6	6.9	10.2	14.1	16.9	19.5	18.6	16.3	13.1	9.3	6.9	(24)
(25)		10%	DB	5.6	5.8	7.5	12.1	16.4	19.5	22.1	22.1	18.9	14.3	9.7	7.0	(25)
(26)			MCWB	4.6	4.6	5.7	8.8	13.2	16.1	18.6	18.3	15.6	12.5	8.6	6.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	8.8	10.6	16.1	18.0	19.8	22.2	21.8	19.2	15.7	11.6	10.0	(27)
(28)			MCD B	10.1	10.1	12.9	23.4	25.0	24.0	27.0	25.9	22.1	17.9	12.5	10.8	(28)
(29)		2%	WB	7.2	7.1	8.6	12.5	16.4	18.5	21.0	20.5	17.8	14.4	10.3	8.4	(29)
(30)			MCD B	8.2	8.5	11.0	17.4	21.6	22.9	24.8	23.8	20.6	16.2	11.3	9.3	(30)
(31)		5%	WB	5.9	5.7	7.0	10.6	14.7	17.4	20.0	19.5	16.9	13.5	9.5	7.0	(31)
(32)			MCD B	6.7	7.0	8.9	14.1	18.2	20.7	23.0	22.5	19.4	15.1	10.5	7.8	(32)
(33)		10%	WB	4.7	4.7	5.9	9.2	13.4	16.4	19.0	18.7	16.1	12.6	8.6	6.1	(33)
(34)			MCD B	5.6	5.8	7.3	11.8	16.0	19.2	21.6	21.5	18.8	14.2	9.6	6.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.6	4.1	4.8	6.0	6.3	6.1	6.2	6.6	6.5	5.2	3.9	3.5	(35)
(36)			MCDBR	4.2	5.6	8.0	11.2	11.3	10.0	10.0	9.9	9.8	7.0	4.5	3.9	(36)
(37)			MCWBR	3.7	4.2	5.2	6.2	5.9	5.7	5.4	5.1	5.3	4.5	3.5	3.5	(37)
(38)		5% WB	MCD BR	4.0	5.0	7.0	9.5	9.8	8.7	8.4	8.2	7.7	6.3	4.3	3.7	(38)
(39)			MCWBR	3.7	4.1	4.9	5.8	5.8	5.6	5.0	5.0	5.0	4.5	3.6	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.341	0.366	0.405	0.401	0.395	0.420	0.417	0.379	0.363	0.345	0.311	(40)	
(41)		taud	2.298	2.304	2.288	2.245	2.318	2.367	2.316	2.326	2.407	2.410	2.331	2.289	(41)	
(42)		Ebn at Noon	619	735	802	817	844	853	821	798	784	705	576	537	(42)	
(43)		Edh at Noon	58	80	101	121	118	114	118	111	90	72	57	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.46	1.10	2.36	4.16	5.44	5.78	5.22	4.33	3.09	1.55	0.61	0.33	(44)	
(45)		RadStd	0.06	0.15	0.23	0.48	0.59	0.42	0.57	0.41	0.33	0.23	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (32 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-130	+60	N/A	

Nomenclature: See separate page